EXAM PAPERS PRACTICE

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Shows that the set is closed under the operation * (must show that under modulo 6, $a * b$ can only result in a member of the given set)	AO2.1	R1	As all answers to $a * b$ are reduced modulo 6 they are in the given set and thus the set is closed under *.
	Clearly identifies the identity element	AO1.1b	B1	Identity element = 2
	Finds and states the inverse of each element (PI) FT from 'their' identity	AO1.1b	B1	0 and 4 are inverses of each other 1 and 3 are inverses of each other 2 and 5 are self-inverse elements
	Shows associativity between elements of the set under the operation *	AO2.1	R1	$a * (b * c) = a + (b + c + 4) + 4$ is shown to equal $(a * b) * c = (a + b + 4) + c + 4$
	States correct conclusion that G is a group under the operation * Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	As G satisfies each of the four group axioms under the binary operation *, G is a group
(b)	Identifies two correct subgroups. Condone inclusion of {0,1,2,3,4,5}	AO1.1b	B1	{2}, {0, 2, 4}, {2, 5}
	Identifies all three proper subgroups and no others included	AO1.1b	B1	

(c)	Identifies the generator of G	AO3.1a	B1	$G = (\langle 1 \rangle, *)$ OR

OR generates every element of the group K (PI)			$K = (\{3,9,13,11,5,1\}, \times_{14})$
Finds correctly a one-to-one mapping between each element of G and K (condone use of equal sign for this mark)	AO1.1b	B1	$\begin{array}{ccc} G & & K \\ 1 & \mapsto & 3 \\ 0 & \mapsto & 9 \\ 5 & \mapsto & 13 \\ 4 & \mapsto & 11 \\ 3 & \mapsto & 5 \\ 2 & \mapsto & 1 \end{array}$
Deduces that G is isomorphic to K with a concluding statement using the correct mathematical language, having used the correct notation throughout.	AO2.2a	E1	As there is a one-to-one mapping between the elements of G and the elements of K, $G \cong K$
			Total 10 marks